

Maths Reasoning

Maths reasoning is all about applying logical and critical thinking to mathematical problems. It involves finding the best possible solutions to mathematical problems and choosing the correct/most efficient methods of solving an issue. It aims to test your child's constructive thinking skills as well as mathematical understanding.

These types of problems can involve:

Code Breaking - Breaking a code where each letter of the alphabet is represented by a number or symbol.

Missing Number Problems - Using different operations to find the missing numbers in calculations.

Algebra - Using orders of calculations to solve problems where letters are used to represent numbers.

Data Handling - Solving problems around interpreting and using data like graphs or timetables.

Here is an example of the type of problem you might come across through maths reasoning:

Given the rule $2n + 3$, what would the first 5 terms of this sequence be?

What would the 15th term be?

Answers:

$$(2 \times 1) + 3 = 5$$

$$(2 \times 2) + 3 = 7$$

$$(2 \times 3) + 3 = 9$$

$$(2 \times 4) + 3 = 11$$

$$(2 \times 5) + 3 = 13$$

$$15\text{th term} = (2 \times 15) + 3 = 33$$

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<https://www.twinkl.co.uk/resource/t2-pa-91-11-survival-guide-for-parents-carers-guide>

11+ Maths Reasoning Practise Paper

Complete the Calculation 3

In the following questions, you have to find the number that will complete the calculation correctly. To complete the calculations, you will need to use all four operations of addition, subtraction, multiplication and division. Speed is of the essence with these questions, but remember to always double-check the calculation to ensure you have put the correct answer!

Example:

$$32 + 4 = 6 \times \square$$

- a. 5
- b. 6
- c. 7
- d. 8

Answer: 6

This is because $32 + 4 = 36$ and $6 \times 6 = 36$.

1. Find the number that will complete the calculation below.

$$12 \times 4 = 32 + \square$$

- a. 14
- b. 16
- c. 18
- d. 20

2. Find the number that will complete the calculation below.

$$143 + \square = 200$$

- a. 75
- b. 67
- c. 57
- d. 76

3. Find the number that will complete the calculation below.

$$72 \div 8 = \square - 15$$

- a. 24
- b. 34
- c. 14
- d. 15

4. Find the number that will complete the calculation below.

$$49 - \square = 5 \times 7$$

- a. 12
- b. 13
- c. 14
- d. 15

5. Find the number that will complete the calculation below.

$$29 + 27 = 7 \times \square$$

- a. 5
- b. 6
- c. 7
- d. 8

6. Find the number that will complete the calculation below.

$$9^2 = 65 + \square$$

- a. 16
- b. 17
- c. 25
- d. 32

7. Find the number that will complete the calculation below.

$$27 - \square = 2 \times 7$$

- a. 12
- b. 13
- c. 14
- d. 15

11+ Maths Reasoning Practice: Complete the Calculation 3 **Answers**

1. 16

2. 57

3. 24

4. 14

5. 8

6. 16

7. 13

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